PROF. DR.-ING. MARK SCHUTERA

PAPERS OVER TIME

UNFINISHED LECTURE NOTES

published by unfinished lecture notes

These notes are, by their very nature, unfinished, and they improve with every reader. If you spot an error, disagree with a framing, or want to add a source, a question, or a position, open a pull request at https://github.com/Quillstacks/LectureMaterial/tree/main/lecturenotes/notes_papersovertime. Every contribution is welcome.

2026-05-08 · local draft

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Introduction

PAPERS OVER TIME reads the field through its primary literature. Each session lifts a single paper, situates it in its moment, and asks what it changed and what it left unfinished. The aim is fluency in the genre and taste in the questions.

Zebrafish Phenotype Pattern Recognition

2026-05-08 · local draft

Counting Embryos

Automated Phenotype Pattern Recognition of Zebrafish for High-throughput Screening
Schutera et al. (2016)
Bioengineered 7(4)

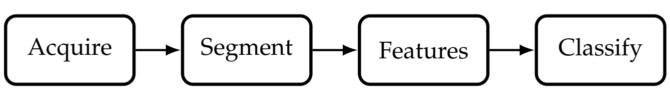
A single screen produces thousands of embryos. Each one has to be inspected at fixed time points to decide which ones carry the phenotype of interest. By hand, the screen never finishes.

Imaging had outpaced scoring. Embryos could be photographed faster than a biologist could classify them. The bottleneck had moved.

Bachelor thesis. The first paper, written before any of this was supposed to be a career.

Question	Can a classifier replace the human at the microscope?
Setting	Four phenotypes at 72 hpf, reduced to wild type vs. variant.
Headline	79% to 99% accuracy, no user interaction.

Model organism. Zebrafish (*Danio rerio*) embryos are transparent, cheap, and develop fast. The phenotype is visible at 72 hours post fertilization.



The move. Treat acquisition and classification as one system. The features the classifier needs are the features the rig is set up to capture.

No deep learning. The data set is small, the phenotypes are well described by intensity and geometry, and a domain expert can name the features that matter. Classical pattern recognition ships.

The contribution is the loop. Rig plus software runs unattended on a real screening question and returns numbers a biologist can act on. Algorithm second, system first.

Take it from here. Drift across screens, rigs, and batches becomes the dominant problem. Labels become the next bottleneck. The later papers pick the thread up.